Adriel Ventura

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Education

Hofstra University — Hempstead, NY

B.S. Mechanical Engineering, Aerospace Concentration — *Expected Dec. 2027*

- Rabinowitz Honors College · Provost Scholar · CSTEP Scholar
- Coursework: Heat Transfer, Thermodynamics, Fluid Mechanics, Computer-Aided Design, Engineering Materials

Experience

Stellant Systems, Inc. — May 2026 – Present

Mechanical Engineering Co-op — *Melville, NY*

- Support mentor-guided steady-state thermal analysis of RF and defense electronic assemblies in ANSYS Mechanical: geometry clean-up, meshing, boundary-condition debugging, and gradient checks against operating limits
- Revised thermal-pad contact conductance in a thermal model to address artificial temperature gradients
- Designed and fabricated a multi-diameter O-ring cutting fixture through four iterations with technician feedback; applied GD&T, tolerance stack-up, and FDM prototypes to address blade deflection and improve cutting repeatability
- Create and audit Bills of Materials and author Engineering Change Notices (ECNs) for production hardware, routed for cross-functional approval through SharePoint

Projects

CPU Heat Sink Thermal Design | *ENGG 114 Heat Transfer; MATLAB, Python, Fusion 360* — Spring 2026

- Built a MATLAB/Python thermal-resistance model with laminar Nusselt correlations: 0.4485 °C/W and 69.85°C; Fusion 360 FEA predicted 69.775°C
- Chose 15 fins over a cooler 17-fin baseline to widen fin clearance and cut mass 27.5% (75.3 g to 54.6 g); wrote a thermal-test plan (not yet experimentally verified)

NASA L'SPACE Mission Concept Academy | *Thermal Engineer, Team 6 (Artemis)* — Aug. 2026 – Present

- Defined lunar-rover thermal requirements, including a −133 to +121 °C operating envelope (requirement SYS-10)
- Evaluated multilayer insulation, surface coatings, and conductive thermal straps for avionics and payload thermal control; added lunar-dust research (NASA TFAWS 2024) to thermal-risk trade studies

Leadership

Society of Hispanic Professional Engineers (SHPE), Hofstra — May 2026 – Present

President & Chapter Revamp Lead — *Hempstead, NY*

- Leading revival of a chapter inactive since 2007: recruited an executive board, authored the constitution, filed university recognition through GetInvolvedHU, and held the first Meet the Board event (Oct. 7, 2026)

American Society of Mechanical Engineers (ASME), Hofstra — May 2025 – Present

Co-President & Social Media Manager (Apr. 2026–); Recruitment Chair (May 2025–Apr. 2026) — *Hempstead, NY*

- Co-lead the executive board; organized steam-turbine demos, catapult builds, a multi-org STEM game night, and two networking events connecting 20+ students with industry mentors

Honors & Certifications

1 of 27 participants advanced to the live SHPE x Rockwell Automation AI Hackathon (2026)
Winner, SHPE's Got Talent, Hofstra SHPE (2026)
$1,350 in Hofstra travel awards (Provost, Equity & Inclusion, Dean's Office) to compete at SHPE 2026

Technical Skills

Analysis: ANSYS Mechanical (steady-state thermal, meshing), MATLAB, Python, Excel macros, thermal-resistance modeling
CAD: Autodesk Inventor, Siemens NX, SOLIDWORKS, Fusion 360, AutoCAD
Manufacturing: GD&T (ASME Y14.5), tolerance stack-up, DFMA, fixture design, FDM/SLA 3D printing
Documentation: Engineering drawings, BOM creation and audit, ECNs, SharePoint
Data & AI: Databricks, AI-assisted development (Claude), Git